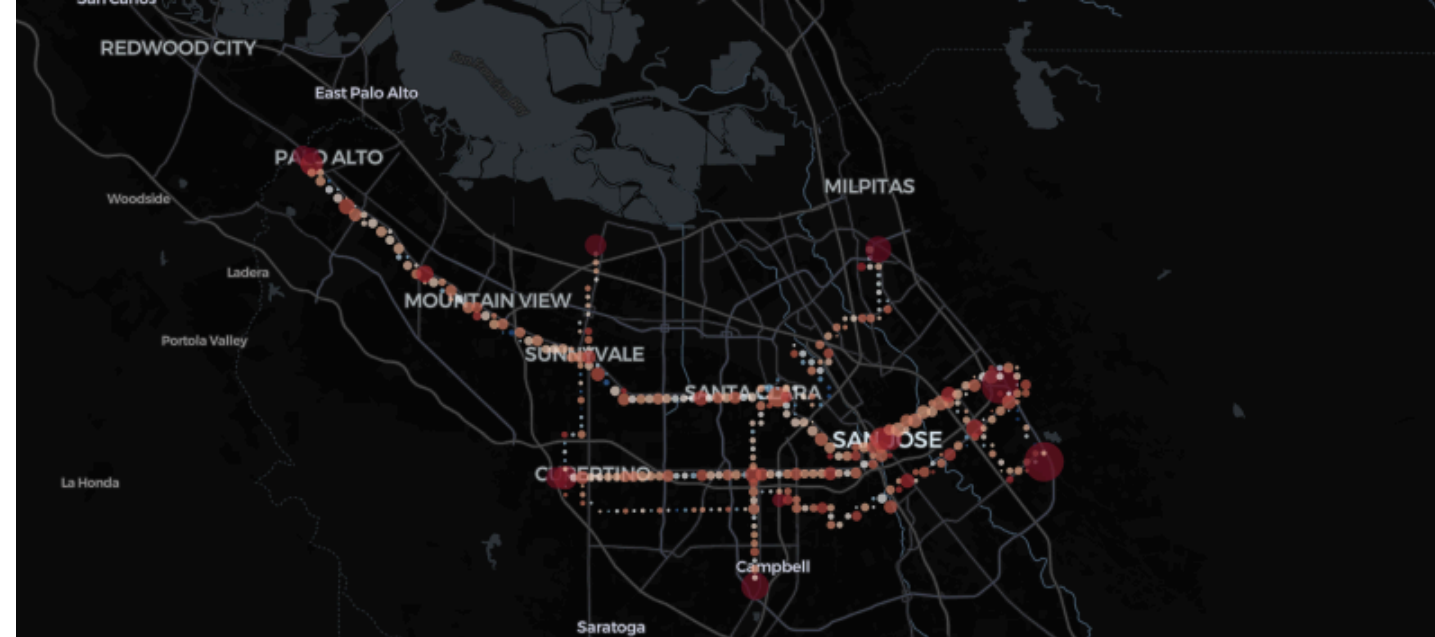




Stop Spacing, Ridership, and Delay on VTA Routes

CYPLAN 255

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May 12, 2026

Introduction and Motivations

Public transit reliability is a fundamental equity issue. When buses run off schedule, riders miss connections, arrive late to work, and lose trust in the system. Research has shown that unreliable bus service disproportionately affects low-income riders who depend on transit as their primary mode of transportation, and that schedule adherence is among the top factors influencing rider satisfaction and retention (Walker, 2012). In the United States, the Federal Transit Administration estimates that bus on-time performance averages around 70–75% systemwide, meaning roughly one in four bus trips deviates significantly from its published schedule (FTA National Transit Database, 2023). In Santa Clara County, the Valley Transportation Authority (VTA) operates a network of high-frequency bus corridors serving dense employment and residential areas. Yet reliability along these corridors varies considerably, and the operational factors associated with delay patterns remain underexplored at the stop level.

This project investigates whether two observable service characteristics – stop spacing (the distance between consecutive stops) and stop-level passenger demand (average daily boardings) – are associated with arrival delay (the deviation between observed and scheduled arrival times) along six VTA frequent-service routes: Routes 22, Rapid 522, 23, Rapid 523, 25, and 60. We also examine whether delay hotspots spatially overlap with high road traffic volumes, using San José Average Daily Traffic (ADT) data as a contextual variable.

Our findings are intended to support VTA operations and service planning staff in identifying priority stops and segments for targeted interventions such as signal priority, stop consolidation, or schedule adjustment. We sought to answer the following research questions:

- **RQ1:** Is shorter stop spacing associated with greater arrival delay at the stop level, controlling for route and time of day?
- **RQ2:** Are stops with higher average daily boardings associated with greater arrival delay, controlling for route?
- **RQ3:** Do delay hotspots (stops in the top quartile of mean delay) spatially overlap with high-traffic road segments as measured by ADT?

Data Description

Our analysis integrates six datasets, each contributing a distinct layer of the corridor picture.

Dataset	Coverage	Granularity
VTA Bus Routes and Stop Geometries	Santa Clara County VTA Bus Routes (selected frequent routes)	Polyline Geometries by route_id and Point Geometries by stop_id
VTA Stop-Level Ridership (2025)	VTA bus stops/corridors, Santa Clara County, 2025	Aggregated Stop-level, Average Daily
City of San Jose Average Daily Traffic*	City of San José Roads	Aggregated Average Daily Point-level
GTFS Realtime-Derived Vehicle Positions, Directions, and Speeds	selected VTA bus routes on Thursday March 26, 2026, ~10:30–18:30.	Timestamped Point Geometries, updated approximately every 30 seconds
GTFS Realtime Trip Updates Delays	actual trips on selected VTA bus routes on Monday, May 4, 2026, ~10:30–18:30.	Timestamped Estimated Bus Arrivals Times at stop_ids (seconds)
VTA GTFS Static Feed Data	Scheduled trips, stops, stations, and routes on selected VTA bus routes	Scheduled Stop Timings by Trip ID (seconds), Station Point Geometries

Table 1: A listing of the datasets used in this study. *ADT data for Santa Clara County was unavailable. **The data was queried using Python code, periodically extracting live responses from the 511 GTFS API.

Type	Format	Source
Discrete Numerical Data	Geospatiao (GeoJSON), Tabular (CSV)	VTA Open Data
Continuous Numerical Data (ridership averages)	Tabular (.xlsx)	VTA Open Data
Continuous Numerical Data (traffic averages)	Tabular (CSV)	San Jose GIS Open Data
Discrete Numerical (vehicle position), Nominal Categorical Data (vehicle travel direction), and Continuous Numerical (vehicle speed and time) data	Tabular (CSV generated from queried GTFS-realtime data (bus_id, trip_id, route_id, lat, lon, timestamp, speed, direction))	511 GTFS Realtime Vehicle Positions Feed**
Continuous Numerical Data (estimated station arrival times)	Tabular (CSV generated from queried GTFS-realtime data (trip_id, route_id, direction_id, start_time, start_date, timestamp, stop_sequence, stop_id, arr_time, dep_time))	511 GTFS Realtime Trip Updates Feed**
Continuous Numerical Data (Scheduled Stop Times), Discrete Numerical Data (Station Locations)	Tabular (comma separated .txt files)	VTA GTFS Data

Table 1: A listing of the datasets used in this study. *ADT data for Santa Clara County was unavailable. **The data was queried using Python code, periodically extracting live responses from the 511 GTFS API.

Route geometries and stop locations provide a spatial skeleton, while ridership data provides stop-level demand. GTFS Static and TripUpdates data allow for delay computation using the differences between scheduled and actual station arrival times. ADT data provides road traffic context to help evaluate its impact on bus delays, though its coverage is limited to San José city boundaries. We were unable to obtain ADT data for Santa Clara County as a whole.

Data Cleaning

Our cleaning pipeline follows 8 reproducible steps, each documented in 02_data_cleaning.ipynb.

- 1. Route and Stop Geometries:** We loaded the October 2025 VTA route shapes (GeoJSON), reprojected all geometries to California State Plane Zone 3 (EPSG:2227) for accurate distance measurement, and filtered for our six study routes. Column names were standardized, and a spatial plot was generated to verify correct route selection.
- 2. Stop-Level Ridership:** We joined the stop location file with the ridership Excel file by station ID, standardized route ID formats (separating multi-route stops into individual records), added a projected geometry column, filtered for study routes, and removed duplicates. The output is a clean table of 654 stop-route records with boardings, alightings, and geometry.
- 3. San José ADT:** We removed unused columns, standardized road names to uppercase, and buffered our route corridors by 250 feet. A spatial join filtered ADT counters within this buffer, retaining only traffic counts directly adjacent to bus corridors. We then used DBSCAN clustering (radius = 350 ft) to group nearby counters at the same intersection and averaged their

values to reduce noise from multiple counters at a single location.

4. **GTFS Realtime Vehicle Positions:** Raw position records were filtered to remove rows with missing coordinates, speed, or direction. We converted the data to a GeoDataFrame and performed a spatial join with the buffered route corridor to exclude off-route vehicles (e.g., buses deadheading on freeways). Duplicate records with the same vehicle, trip, route, and timestamp were dropped. The cleaned dataset contains approximately 80,000 unique vehicle-position records.

5. **Delay Computation:** We merged data from GTFS TripUpdates (actual arrival times) with GTFS Static scheduled stop times, joining on trip_id and stop_sequence. Times were converted to seconds from midnight, with a midnight-crossing correction for late-night trips. Delay was computed as actual_arrival_sec – scheduled_arrival_sec. Negative values indicate early arrival; positive values indicate late arrival. Records with missing arrival times were dropped.

6. **Unified Analysis Table:** We merged the delay and ridership datasets on [route_id, direction_id, stop_id] using an inner join.

7. **Stop Spacings:** We computed stop spacings using OpenStreetMap network routing (via OSMnx) between consecutive stops along each route. We found the nearest node to each station, since stations tend to be located at intersections, dropping any null spacings. The final unified table contains 710 stop-route records.

8. **ADT Analysis Table for San Jose Stations:** We created an additional, separate dataset for stations within the City of San Jose, using DBSCAN to cluster nearby ADT counters and using proximity to nearby stations to join stations to their closest cluster of ADT counters. This set contains 255 stop-route records.

Methods of Analysis

Our analytical approach combines descriptive statistics, OLS regression, and spatial visualization to characterize the

associations between stop spacing, passenger demand, and arrival delay. Our unit of analysis was at the stop-route level; that is, each row in our analysis table represents one stop on one route, in one direction. This allows us to compare stops across routes while accounting for route-level differences.

Independent Variables (X):

- Stop Spacing (spacing_ft): Distance in feet between a stop and the preceding stop along the route, computed via OSMnx shortest-path routing on the OpenStreetMap road network
- Boardings: Average daily boardings at the stop (2025 ridership data)
- Traffic Counts (ADT): Average daily vehicle traffic at nearby road intersections (San José only; Model 2 only)

Dependent Variable (Y):

- mean_delay: Mean arrival delay in seconds across all observed trips at that stop on May 4, 2026. Positive = late; negative = early.

We produced the following OLS Regression models using our analysis tables:

- **Model 1 (all 710 stops):** `mean_delay ~ boardings + spacing_ft`
- **Model 2 (255 San José stops with ADT coverage):** `mean_delay ~ boardings + spacing_ft + adt`

Both models standardize independent variables using z-score scaling to allow coefficient comparison. We complemented OLS with Pearson and Spearman rank correlations as a robustness check, since stop spacing has a long right tail (Route 25 contains several freeway-gap segments).

In our spatial analysis, delay hotspots are defined as stops in the top quartile of mean delay (≥ 66 seconds late). We use a Plotly Express bubble map to visualize the joint distribution of ridership

and delay across the corridor network, and a two-layer overlay map to test RQ3 (hotspot–ADT spatial overlap).

Results and Visualizations

Across our 710 stop-route records from May 4, 2026, we find that VTA buses on our selected lines are, on average, running ahead of schedule:

- Mean arrival delay: −28.1 seconds (early)
- Median arrival delay: −33.4 seconds (early)
- Delay hotspot threshold (75th percentile): +66.0 seconds late
- Number of hotspot stops: 178 out of 710 (25%)
- Mean daily boardings across all stops: 77.7 passengers/day
- Mean stop spacing: 452 meters

Splitting arrival delays by route, we saw that while routes 23, 60, 522, and 523 were generally early, routes 22 and 25 were delayed on average.

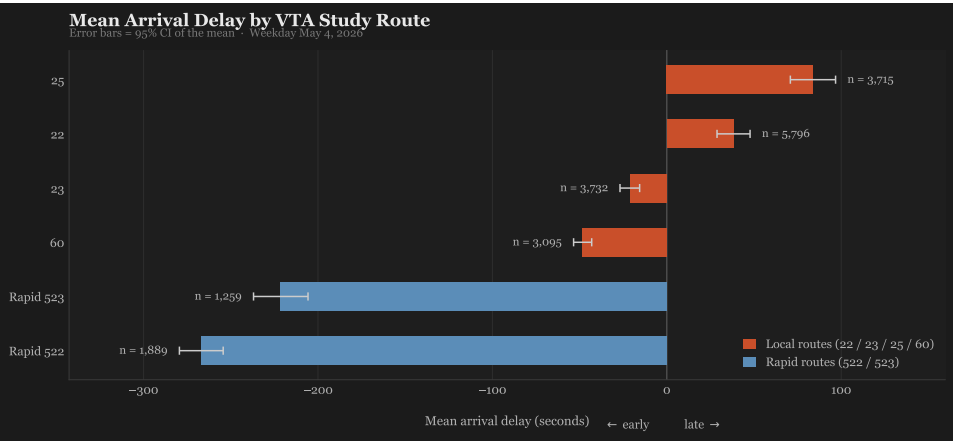


Figure 1: Mean Arrival Delay by Route

We also see that Routes Rapid 522 and Rapid 523 tend to run far earlier than their local counterparts, arriving over 3 minutes early on average. We used a horizontal bar chart to allow for easy and direct length comparison of a single metric across named categories. The horizontal layout helps to avoid angled labels and improves readability.

Next, we look closely at the stations with the highest average delays.

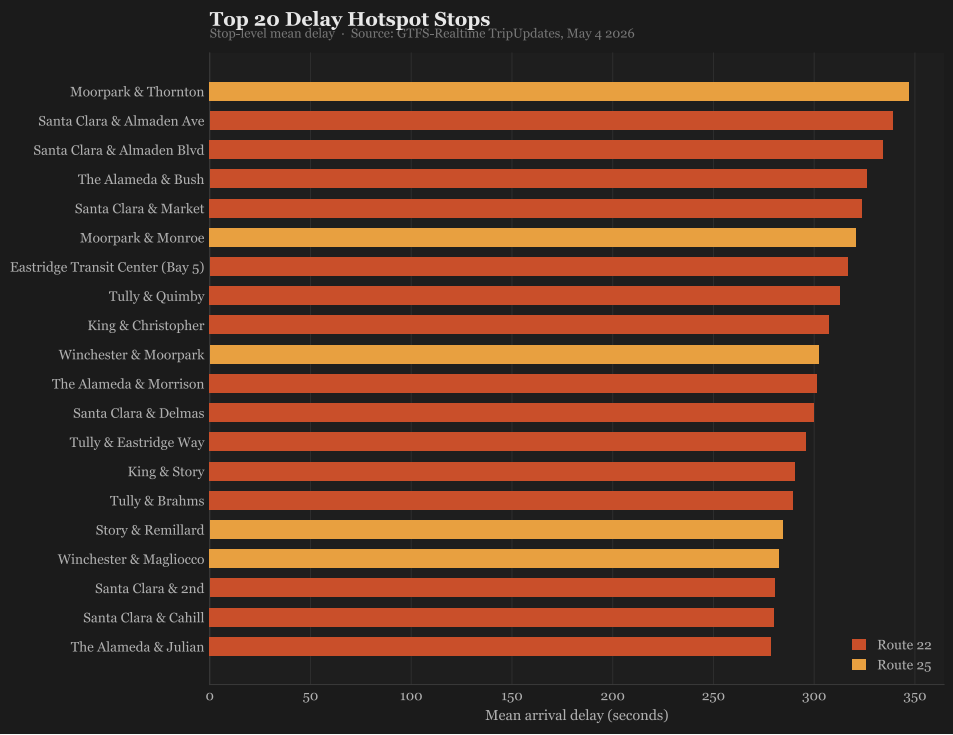


Figure 2: Top 20 Delay Hotspots

Of the 20 stops with the highest mean delay, all of them are on routes 22 and 25. The highest delay is approximately 6 minutes, at Moorpark & Thornton. Once again, a sorted horizontal bar chart helps for easy ranking and comparison of station delay magnitudes across many named stops, for which length is an effective visualizer.

Examining the moving speed by route can help us determine bus cruising speeds, highlighting how often buses tend to move quickly or slowly. To do this, we used live vehicle location and speed, and removed points where the bus was stopped (speed at less than 0.5 m/s). This method is also helpful as it ignores out-of-service bus idling at the terminus, during which GTFS continues to send pings.

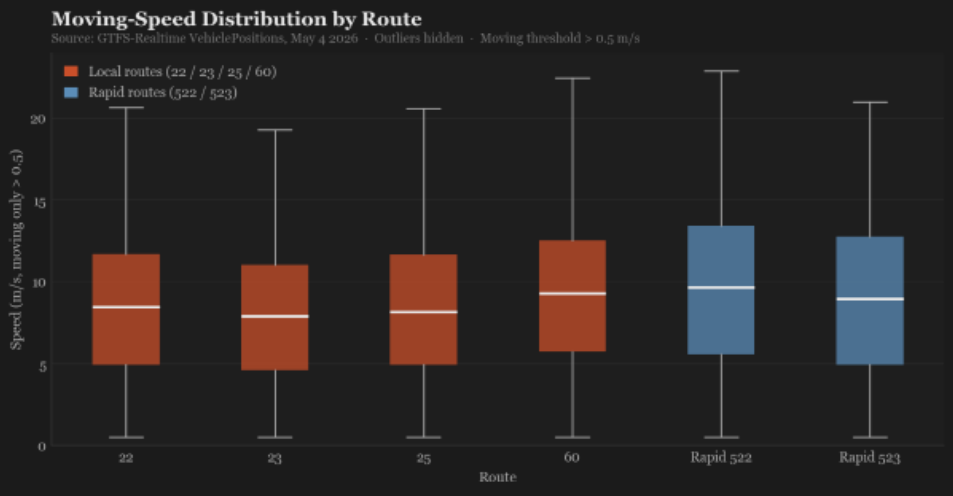


Figure 3: Moving Speed Distribution by Route

Route 23 shows the lowest median speed, while Rapid routes (522, 523) show higher medians, likely consistent with fewer stops and signal priority on some segments. Route 60 also shows some higher values, likely helped by a stretch of roadway near San Jose International Airport, where buses can generally run faster. Box plots help to visualize the greater variability in speeds between routes, giving a more complete picture of operating conditions than a simple mean or median.

We can further examine the share of time each route spends stopped, whether due to station dwelling, stopping at intersections or signals, or other reasons. Unfortunately, due to the aforementioned issue of GTFS continuing to record data for out-of-service buses that are stopped or idling at bus termini, such data can be significantly skewed or inaccurate.

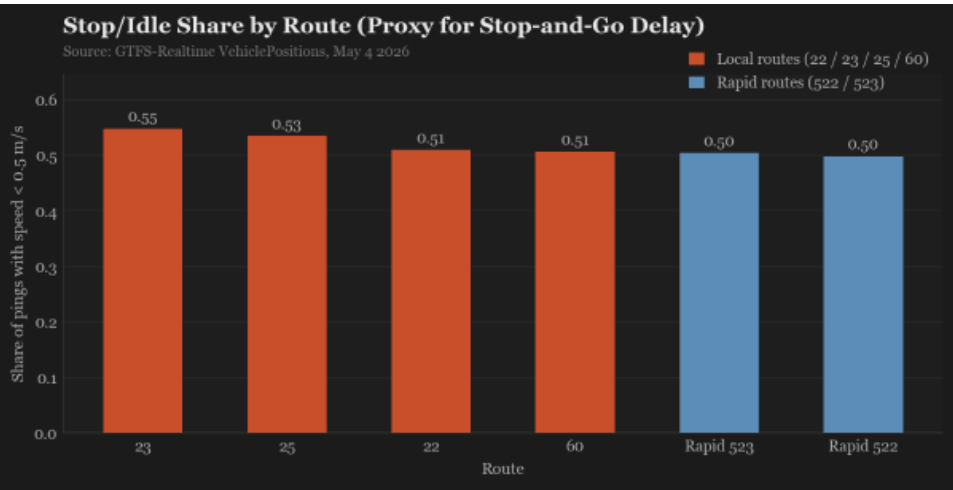


Figure 4: Percentage of travel time stopped (speed < 0.5 m/s)

The Rapid routes have the least idling time, but again, this figure should be taken with terminal dwelling in mind, which can significantly skew actual figures. Once again, a bar chart is effective for comparing a single metric across six routes, allowing for comparison.

Below, we visualized stop spacings along the selected routes, splitting among the local and rapid routes to reduce route overlap and allow for better visualization:

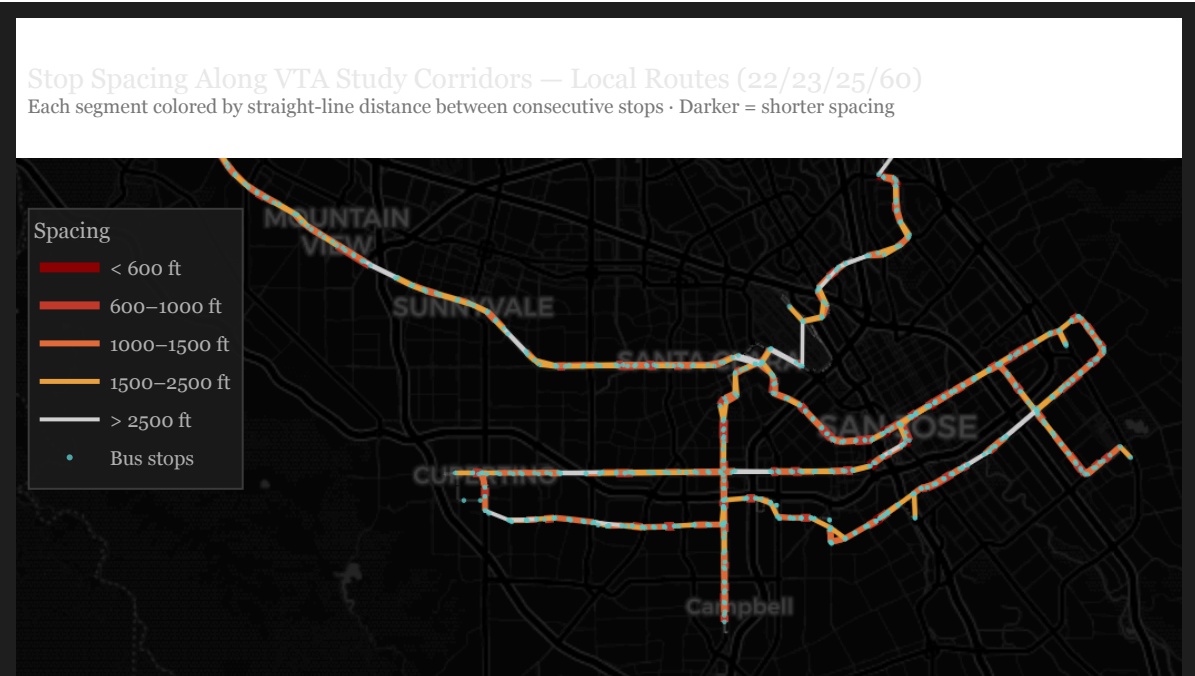


Figure 5a: Geographical Station Spacings for Selected Local Bus Lines (22, 23, 25, 60)

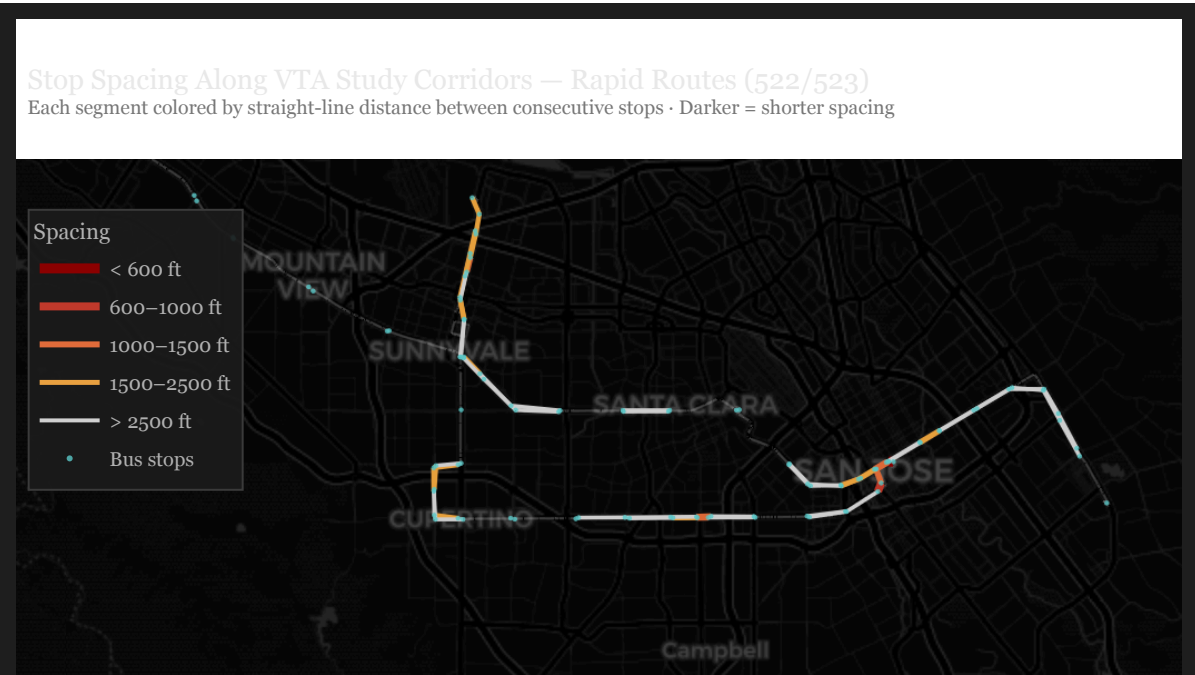


Figure 5b: Geographical Station Spacings for Selected Rapid Bus Lines (522, 523)

In the maps above, dark red indicates a tight station spacing (<600ft), while greys indicate wider stop spacings (>2500ft) . We see the densest station clustering in downtown San Jose and Santa Clara, which is perhaps unsurprising, and the widest spacings on the Rapid bus routes and near freeway interchanges. A mapbox of discrete color encoding helps to categorize the spacing groups along a linear network, preserving the spatial arrangement of stops while effectively highlighting potential areas of interest or concern.

Below, we can see overall low-speed hotspots and the percentage of stopped pings from GTFS, aggregated by 1200-foot cells. While again, this visualization should be disregarded for terminal points on each route, it does provide an effective proxy for both service frequency and areas of frequent slowing outside of those areas.

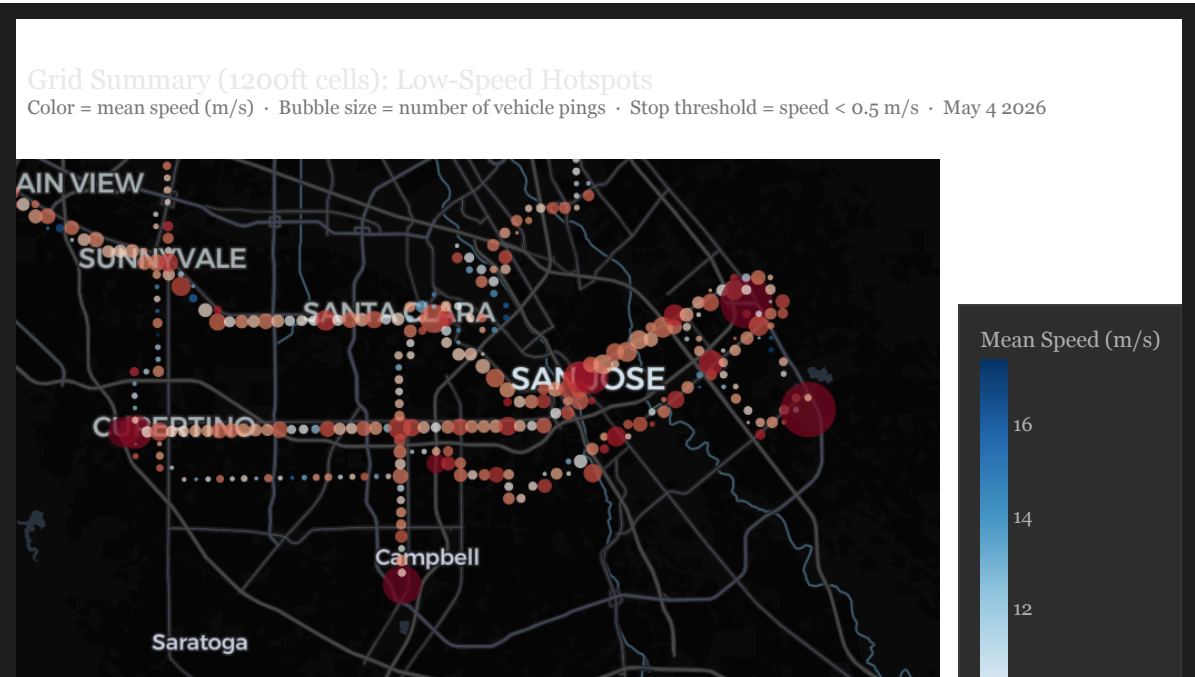


Figure 6: Stop/Idle Share Grid and Stopping Hotspots

In the visualization above, the bubble size encodes the share of near-stopped pings (speed < 0.5 mph), with color indicating mean speed within the cell. The most prominent low-speed clusters again appear in downtown San José. We aggregate to a grid rather than plotting raw GPS points to better handle the ~80,000 records in our data. The 1,200-ft cell size balances spatial resolution against visual clarity.

The results of our OLS regressions are shown below:

Variable	Coefficient	p-value
Boardings	-22.24	< 0.001
Stop Spacing (ft)	-9.38	0.111
R^2	0.023	-

Table 2: OLS Regression Results for Model 1: all selected routes

Significance
significant inverse correlation
Not very significant
-

Table 2: OLS Regression Results for Model 1: all selected routes

Variable	Coefficient	p-value
Boardings	-29.40	0.003
Stop Spacing (ft)	-17.68	0.072
ADT	16.17	0.099
R^2	0.057	-

Table 3: OLS Regression Results for Model 2: all selected stations within the City of San Jose within close proximity of an ADT counter.

Significance
significant inverse correlation
slightly significant inverse correlation
slightly significant direct correlation
-

Table 3: OLS Regression Results for Model 2: all selected stations within the City of San Jose within close proximity of an ADT counter.

These results seem to indicate some inverse correlations between boarding numbers and station spacing to delays, as well as a direct

correlation between ADT and increased delays. These results are better detailed in the visualizations below:

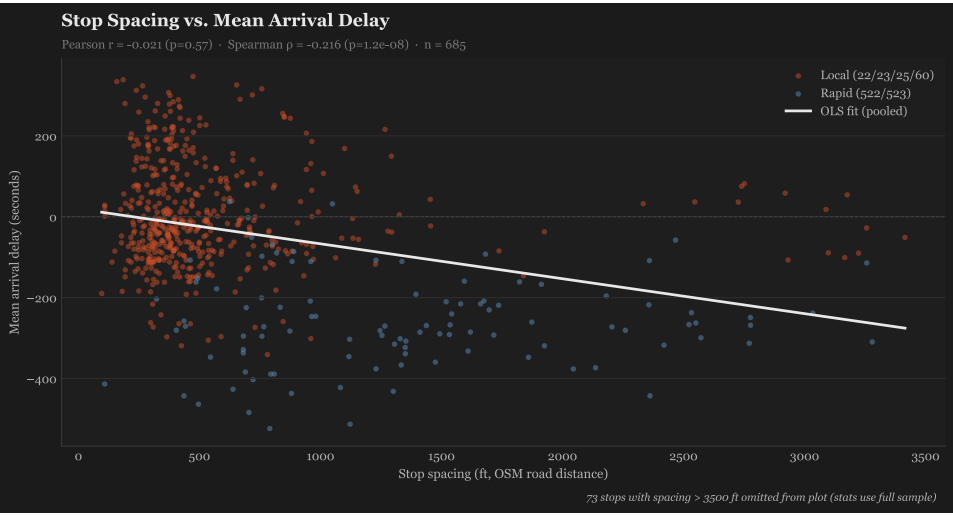


Figure 7: Stop Spacing vs. Mean Arrival Delay

Figure 7 examines RQ1 by plotting stop spacing against mean arrival delay, colored by route type. The pooled Spearman rank correlation is $\rho = -0.216$ ($p < 0.001$), indicating a statistically significant monotonic association: stops with shorter spacing tend to experience greater delays. However, the near-zero Pearson correlation ($r = -0.021$, $p = 0.575$) reveals that this relationship is non-linear, driven largely by two distinct clusters rather than a uniform trend.

Local routes (22/23/25/60) concentrate at shorter spacings (under 1,000 ft) with high delay variance, while Rapid routes (522/523) operate at wider spacings and consistently show negative delays, suggesting they run ahead of schedule. This separation suggests that route type is a confounding factor: the spacing-delay association observed in the pooled sample may partly reflect structural differences between local and rapid service rather than a direct effect of stop spacing alone. The OLS fit line, while visually negative, should therefore be interpreted with caution as a pooled average across two operationally distinct groups.

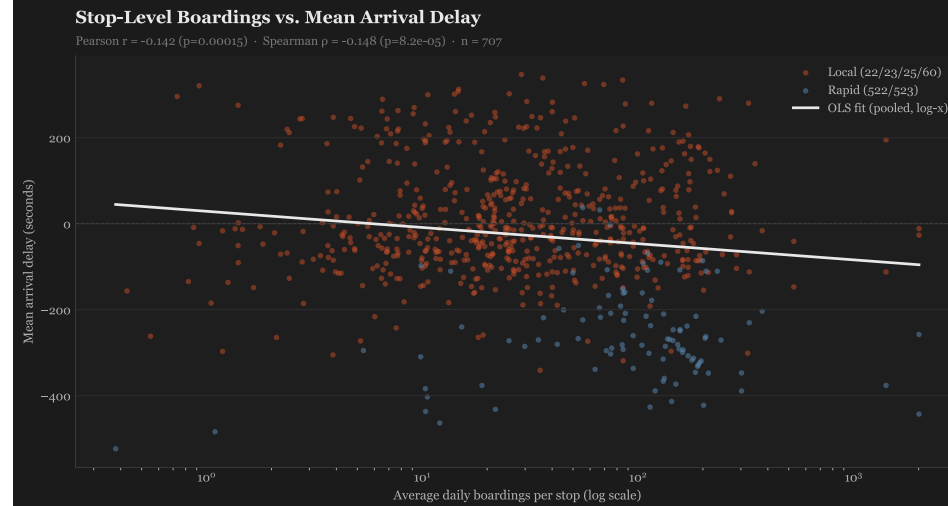


Figure 8: Stop-Level Boardings vs. Mean Arrival Delay

Figure 8 examines RQ2 by plotting average daily boardings against mean arrival delay on a log scale. Both the Pearson ($r = -0.142$, $p < 0.001$) and Spearman ($\rho = -0.148$, $p < 0.001$) correlations are statistically significant, indicating a consistent negative association: stops with higher ridership tend to show less positive (more negative) delay.

This direction is counterintuitive. One might expect busier stops to accumulate greater delays due to longer dwell times; instead, the pattern is reversed. As with Figure 7 above, route type is a likely confounding factor: Rapid 522/523 stops dominate the high-boardings range (around 10^2) and systematically arrive ahead of schedule, pulling the overall correlation negative. Local route stops, by contrast, scatter across lower ridership levels with much higher delay variance. The pooled OLS fit therefore reflects this structural separation between route types rather than a within-route demand-delay tradeoff. Subgroup analysis by route type would be needed to isolate whether ridership independently predicts delay after controlling for service type.

Figure 9 — Stop-Level Ridership and Arrival Delay
VTA Routes 22/522/23/523/25/60 · Weekday · May 2026 · Bubble size = avg daily boardings · Blue = Early · Red = Late

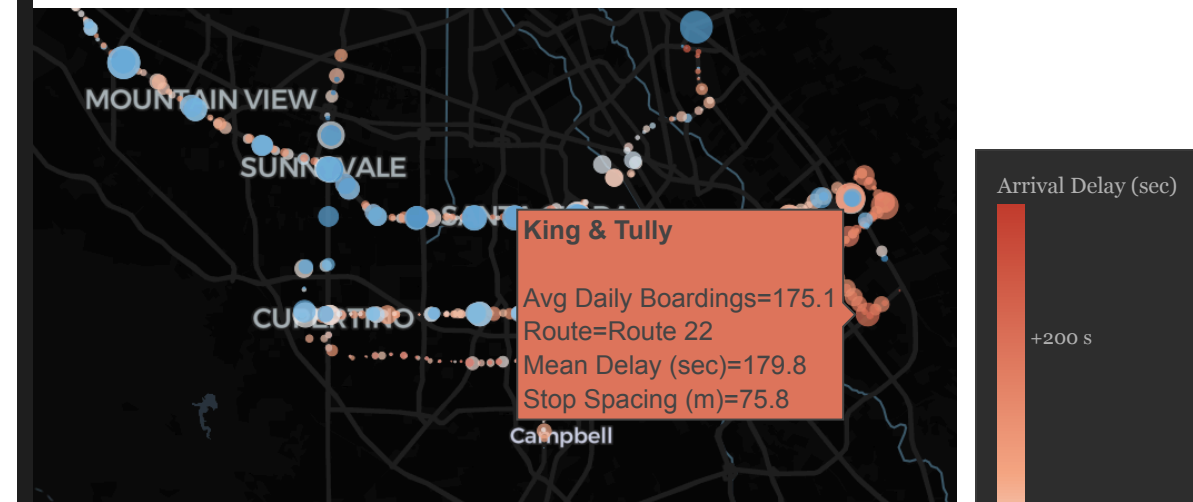


Figure 9: Geographical Stop-Level Ridership and Arrival Delay

Figure 9, rendered in Plotly Express, represents each bus stop with a bubble. Bubble size encodes average daily boardings, and color encodes mean arrival delay (red indicates lateness while blue indicates earliness). The largest bubbles — concentrated on Rapid 522/523 corridors — are predominantly blue, confirming that high-ridership stops tend to arrive early. Red hotspots generally tend to be smaller, suggesting that chronically late stops are not necessarily the busiest ones. We chose to include both metrics in a single bubble map to allow the reader to simultaneously analyze both dimensions (demand and delay).

Finally, the last figure addresses the impacts of ADT on bus delays.

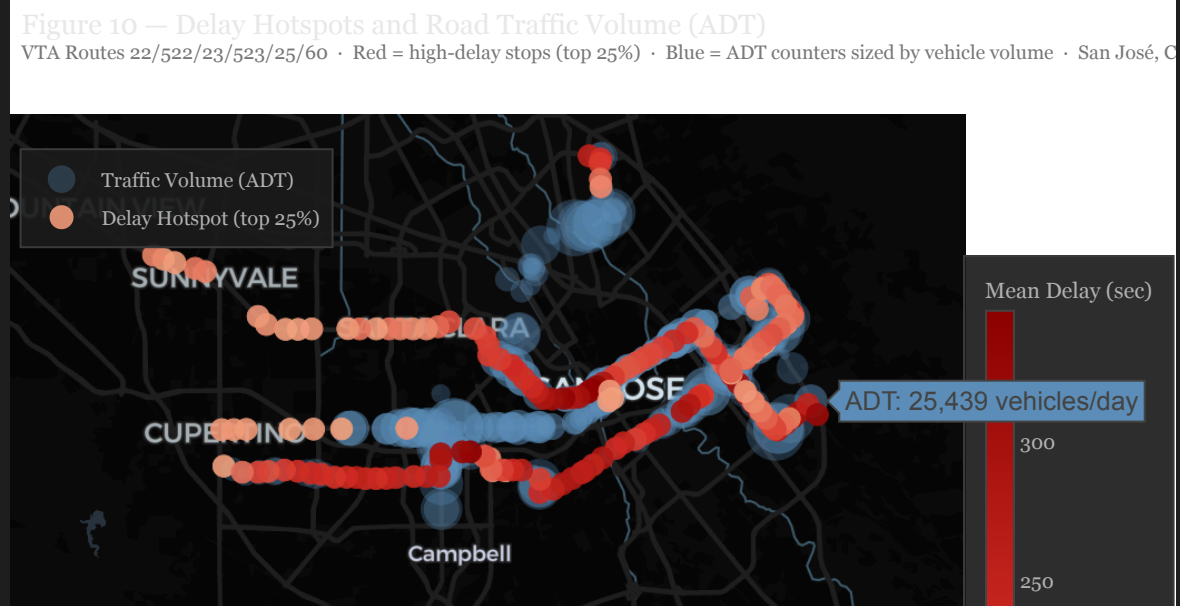


Figure 10 — Overlaid Delay Hotspots and Road Traffic Volume (ADT)

Figure 10 overlays delay hotspot stops (top 25% by mean arrival delay, shown in red) with San José ADT traffic counters filtered to corridors within the study area (grey, sized by vehicle volume). The Spearman correlation between counter-level ADT and stop-level delay is $\rho = +0.173$ ($p = 0.006$), indicating that higher road traffic volumes are positively associated with greater bus delay in the San José subsample — a direction that aligns with expectation.

Spatially, several hotspot clusters concentrate along El Camino Real and Stevens Creek Boulevard, where ADT counters also record some of the highest volumes, suggesting that arterial traffic load may be one factor co-varying with delay in these segments. That said, the ADT coverage is limited to San José and represents a single point-in-time average rather than time-of-day volumes, so the correlation should be interpreted cautiously. A finer-grained analysis matching ADT to delay by time period would be needed to assess whether the association holds across peak and off-peak conditions.

Limitations

This study faces a number of key limitations which would likely require further steps to address for better reliability:

- 1. Sample Size: All speeds and delay metrics are derived from one weekday each of GTFS Realtime data (Thursday, March 26, 2026, for bus locations and speeds, and Monday, May 4, 2026, for trip updates and delays). A single day may not represent typical weekday conditions and can be skewed by temporary disruptions such as atypical traffic, accidents, service patterns, or cancellations. Multi-day replication across varied weekdays would substantially strengthen the robustness of our findings, but would require significant longer term data collection and storage.
- 2. The Modifiable Areal Unit Problem (MAUP): We conduct our primary analysis at the stop level, motivated by data availability (ridership and delay are both stop-level measures) and operational relevance (stops are the natural intervention unit for service planners). However, aggregating to a different unit, such as corridor segments, grid cells, or census tracts, would likely yield different correlation magnitudes. Our grid-level speed maps confirm that spatial patterns are detectable at multiple scales, but we caution against assuming that stop-level findings translate directly to segment- or corridor-level conclusions.
- 3. Ecological Fallacy: Our findings describe aggregate stop-level associations, but we cannot draw conclusions about individual riders' experiences from these aggregate patterns. For example, a stop with high mean boardings and low mean delay does not imply that every rider at that stop experiences on-time service, and individual trip variation may be high even when the stop average appears favorable.
- 4. Dark Data: We identify three categories of missing data using the Dark Data framework:
 - Data not recorded: Stop dwell time is not directly observable from vehicle position pings. Dwell time is a key mechanism linking boardings to delay, but we can only infer it indirectly through delay measurements. Additionally, delay measurements are also estimated, and can be thrown off in real time by unexpected events such as sudden road closures or accidents.
 - Data not accessible: Multi-day GTFS Realtime archives are not available through the 511 API, and no historical data for bus

speeds, positions, and delays were publicly available. These data collection difficulties made single-day observation the best option available to us.

- Data present but not integrated: ADT data is available only for San José city roads, and from the CalTrans data portal for state-maintained routes. The absence of ADT coverage for the remainder of Santa Clara County, including Palo Alto, Sunnyvale, Santa Clara, and Milpitas, results in minimal explanatory power for these regions and a more limited overall picture. As seen in our results, only 255 of 710 stops (36%) in our study have accompanying ADT data. Another example is the case of terminal bus dwelling, where out-of-service buses at the terminal still continue to provide data to GTFS. These are difficult to remove due to being mixed with data from in-service buses, and due to being on the terminal, they cannot be filtered out with geographical buffering either. As such, the unstructured nature of this data results in some of our visualizations being potentially misleading.
- 5. Non-Causal Interpretation: It is critical to emphasize that the relationships discussed above are not necessarily causal, but only correlational. Our OLS models control for observable variables but cannot account for unobserved confounders such as signal timing, intersection geometry, operator behavior, or special events.

Conclusions and Next Steps

This project examines stop spacing, passenger demand, and arrival delay on six VTA frequent-service corridors in Santa Clara County, using GTFS Realtime data from a single weekday in May 2026.

Our central finding is that VTA buses on these corridors are running ahead of schedule on average (mean delay: −28.1 seconds), albeit on a relatively small single-day sample size. Within this context, 178 stops (25%) show chronic positive delay exceeding 66 seconds, with many concentrated along the valley's surface street backbone of El Camino Real, the Alameda, Santa Clara Street, and

Alum Rock Avenue, as well as along the route of the 25 on Willow Street and Williams Road.

Regarding our three research questions:

- RQ1 (stop spacing → delay): Spearman correlation $\rho = -0.22$ ($p < 0.001$) indicates a significant monotonic association – stops with shorter spacing tend to experience greater delay. The OLS coefficient is not statistically significant, likely because of long-tail spacing outliers on Route 25.
- RQ2 (boardings → delay): Higher-ridership stops are unexpectedly associated with less positive delay, likely because the highest-ridership stops are concentrated on Rapid 522/523 corridors that run ahead of conservative schedules.
- RQ3 (hotspot–ADT overlap): Delay hotspots show partial spatial overlap with high-ADT road segments in San José, particularly along Story Road and Alum Rock Avenue, consistent with road traffic contributing to bus delay at those locations.

These findings suggest that schedule reliability on these VTA corridors may benefit from targeted investigation at the stops identified as hotspots – particularly where high road traffic volumes co-occur with chronic lateness. Future work should replicate this analysis across multiple weekdays and expand ADT coverage beyond San José city limits to enable corridor-wide traffic comparisons.

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